

Causal vs. Marginal Effects

2026-06-07

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v readr     2.1.5
## v ggplot2    3.5.0      v stringr  1.5.0
## v lubridate  1.9.2      v tibble   3.2.1
## v purrr      1.0.1      v tidyr    1.3.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
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```
estimate_causal_gcomp <- function(n_timepoints = 100,
                                   n_per_time = 50,
                                   a_zx, a_zpz, b_ypz, b_yx,
                                   c_z, c_zp, c_y,
                                   sd_ez, sd_ezp) {

  total_n <- n_timepoints * n_per_time

  # Generate actual exposure values for each timepoint
  # X_t_actual <- rep(0, n_timepoints)
  X_t_actual <- runif(n_timepoints, 0, 3)
  # Timepoint index for aggregation
  timepoint_idx <- rep(1:n_timepoints, each = n_per_time)

  # Generate errors
  eps_z <- rnorm(total_n, mean = 0, sd = sd_ez)
  eps_zp <- rnorm(total_n, mean = 0, sd = sd_ezp)

  # ===== do(X_t = X_t_actual) =====
  X_do0 <- rep(X_t_actual, each = n_per_time)
  Z_do0 <- a_zx * X_do0 + c_z + eps_z
  Zp_do0 <- a_zpz * Z_do0 + c_zp + eps_zp
  linear_pred_do0 <- b_ypz * Zp_do0 + b_yx * X_t_actual + c_y
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p_do0 <- plogis(linear_pred_do0)
Y_ij_do0 <- rbinom(total_n, 1, p_do0)
Y_t_do0 <- tapply(Y_ij_do0, timepoint_idx, sum)
E_Y_do0 <- mean(Y_t_do0)

# ===== do(X_t = X_t_actual + 1) =====
# Shift exposure up by 1 for each timepoint
X_do1 <- rep(X_t_actual + 1, each = n_per_time)
Z_do1 <- a_zx * X_do1 + c_z + eps_z
Zp_do1 <- a_zpz * Z_do1 + c_zp + eps_zp
linear_pred_do1 <- b_yzp * Zp_do1 + b_yx * (X_t_actual + 1) + c_y
p_do1 <- plogis(linear_pred_do1)
Y_ij_do1 <- rbinom(total_n, 1, p_do1)
Y_t_do1 <- tapply(Y_ij_do1, timepoint_idx, sum)
E_Y_do1 <- mean(Y_t_do1)

# Causal effect: average effect of increasing X by 1
causal_effect <- log(E_Y_do1) - log(E_Y_do0)

return(causal_effect)
}

# Parameters
a_zx <- 0.9; a_zpz <- 1.0; b_yzp <- -0.8; b_yx <- 0.5
c_z <- 0.1; c_zp <- 0.3; c_y <- -4
sd_ez <- 0.5; sd_ezp <- 0.4

# Theoretical true total causal effect
alpha <- a_zpz * a_zx
T_true <- b_yx + b_yzp * alpha

# Data generating function
generate_data <- function(n_timepoints = 100, n_per_time = 50) {
  total_n <- n_timepoints * n_per_time
  X_t <- runif(n_timepoints, 0, 3)
  X <- rep(X_t, each = n_per_time)
  timepoint_idx <- rep(1:n_timepoints, each = n_per_time)

  eps_z <- rnorm(total_n, 0, sd_ez)
  eps_zp <- rnorm(total_n, 0, sd_ezp)

  Z <- a_zx * X + c_z + eps_z
  Zp <- a_zpz * Z + c_zp + eps_zp
  linear_pred <- b_yzp * Zp + b_yx * X + c_y
  Y_ij <- rbinom(total_n, 1, plogis(linear_pred))
  Y_t <- tapply(Y_ij, timepoint_idx, sum)

  return(data.frame(timepoint = 1:n_timepoints, X = X_t, Y = Y_t, n = n_per_time))
}

# Estimate marginalized effect
estimate_marginalized <- function(data) {
  fit <- glm(Y ~ X, data = data, family = poisson(), offset = log(n))

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    return(coef(fit)["X"])
}

# Run simulation
run_simulation <- function(n_reps = 200, n_timepoints = 100, n_per_time = 50) {

  results <- data.frame(
    rep = 1:n_reps,
    marginalized_effect = NA_real_,
    causal_effect = NA_real_
  )

  for (rep in 1:n_reps) {
    if (rep %% 50 == 0) cat("Replication", rep, "of", n_reps, "\n")

    # Generate data
    data <- generate_data(n_timepoints, n_per_time)

    # Marginalized effect
    results$marginalized_effect[rep] <- estimate_marginalized(data)

    # Causal effect (using corrected function with x_ref = mean X)
    # x_ref <- mean(data$X)
    results$causal_effect[rep] <- estimate_causal_gcomp(
      n_timepoints = n_timepoints,
      n_per_time = n_per_time,
      # x_ref = x_ref,
      a_zx = a_zx, a_zpz = a_zpz, b_yzp = b_yzp, b_yx = b_yx,
      c_z = c_z, c_zp = c_zp, c_y = c_y,
      sd_ez = sd_ez, sd_ezp = sd_ezp
    )
  }

  return(results)
}

# Run
set.seed(42)
cat("Running simulation...\n")
sim_results <- run_simulation(n_reps = 5000, n_timepoints = 100, n_per_time = 80)

# Summary
summary_stats <- sim_results %>%
  summarise(
    marg_mean = mean(marginalized_effect, na.rm = TRUE),
    marg_sd = sd(marginalized_effect, na.rm = TRUE),
    causal_mean = mean(causal_effect, na.rm = TRUE),
    causal_sd = sd(causal_effect, na.rm = TRUE),
    diff_mean = mean(marginalized_effect - causal_effect, na.rm = TRUE)
  )
summary_stats

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## ===== RESULTS =====
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## Theoretical True EFFECT: -0.22
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## Marginalized mean: -0.2183
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## Causal mean: -0.2171
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## Difference: -0.0012
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## =====
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# Histogram
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sim_results_long <- sim_results %>%  
  pivot_longer(cols = c(marginalized_effect, causal_effect),  
    names_to = "effect_type",  
    values_to = "estimate")  
  
ggplot(sim_results_long, aes(x = estimate, fill = effect_type)) +  
  geom_histogram(alpha = 0.6, bins = 40, position = "identity") +  
  geom_vline(xintercept = T_true, linetype = "dashed") +  
  labs(title = "Causal vs Marginalized Effect",  
    x = "Estimate", y = "Count", fill = "Effect") +  
  theme_minimal()
```

